

# YOUSSEF AHMED KOURA

AI / Machine Learning Engineer & Mechatronics Engineering Graduate  
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github.com/Yousef-koura | linkedin.com/in/yousefkoura | Portfolio: yousefkoura.github.io

*This document is a unified knowledge base compiled for RAG (Retrieval-Augmented Generation) purposes. It merges two CV versions and the portfolio website content into a single searchable source.*

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## 1. Professional Summary

Mechatronics Engineering graduate (GPA 3.62 / 4.0) from MSA University (supervised by University of Greenwich, UK) with hands-on experience building and deploying ML and deep learning models across computer vision, classification, and data pipeline workflows. Published researcher who deployed real-time AI on edge hardware (NVIDIA Jetson Nano) achieving 96% accuracy at 20 FPS. Skilled in Python, PyTorch, TensorFlow, and Docker, with practical exposure to MLOps and cloud-adjacent tools.

Passionate about developing innovative solutions at the intersection of robotics and artificial intelligence. Strong leadership and project management abilities demonstrated through academic group projects and internships. Actively seeking roles in AI/ML engineering, deep learning, or data engineering.

## 2. Education

### ***B.Sc. Mechatronics Systems Engineering***

**MSA University** (supervised by University of Greenwich, UK) — Giza, Egypt

Duration: 2020 – 2024 | GPA: 3.62 / 4.0

Relevant Coursework: Stress Analysis, Computer Programming, Electric Drive System, Sensors, Mechatronics Modelling and Simulation, Project Management, CAD/CAM, Quality Management, Mechanical Vibration, Manufacturing Process, Robotics, Automatic Control, Microprocessor and Microcontroller, Artificial Intelligence, Hydraulic & Pneumatic Control, Industrial Automation.

### ***High School Diploma***

Sharjah Public School | Graduated June 2019 | Grade: 97 / 100

## 3. Work Experience

### ***Machine Learning Engineer (Remote)***

**PioPetro — Ohio, United States** | Jun 2024 – Aug 2024

- Completed a virtual training program focused on ML applications in the petroleum industry.
- Covered domains: Petroleum Exploration, Artificial Lift Technology, Well Intervention, and Well Stimulation.
- Applied supervised learning techniques to petroleum datasets as part of structured ML pipelines.

### ***AI/ML Intern***

**Information Technology Institute (ITI) — Al Minufiyah, Egypt** | Jul 2023 – Sep 2023

- Developed and benchmarked supervised and unsupervised ML models (classification, regression, clustering) using Scikit-learn.
- Applied cross-validation and performance metrics (Accuracy, F1, RMSE) across structured datasets.

- Gained foundational experience in Python-based ML workflows and model evaluation.

### ***AI/ML Intern***

**Digital HUB (D-HUB) — Cairo, Egypt | Aug 2023**

- Built a breast cancer detection classifier achieving 95% accuracy.
- Awarded 3rd place in a competitive AI challenge co-hosted with the Electronics Research Institute.

### ***Engineering Intern***

**Elmarakby Steel | Summer 2022**

- Completed a general training program across all engineering departments including Production, Maintenance, Quality, R&D; Hydraulic, Refractory, Electrical, and Safety.
- Assisted engineers in quality control checks, troubleshooting equipment issues, and analyzing data to improve production processes.

## **4. Projects**

### ***Agri-Bot — AI-Powered Crop Disease Detection and Medication Recommendation Robot***

*Graduation Project | Nov 2023 – Jun 2024 | <https://github.com/Yousef-koura/AGRI-BOT>*

- Designed a real-time computer vision pipeline using YOLOv8 (detection) and MobileNetV2 (classification).
- Deployed on NVIDIA Jetson Nano, achieving 96% accuracy at 20 FPS.
- Implemented fuzzy PID motor control for autonomous navigation.
- Built a web interface for remote monitoring and disease diagnosis.
- Published as a research paper at IUGRC 2024.

### ***FMCG Data Engineering Pipeline***

*Data Engineering | 2025 | [https://github.com/Yousef-koura/FMCG\\_Data\\_Engineer\\_Project](https://github.com/Yousef-koura/FMCG_Data_Engineer_Project)*

- Built an end-to-end Lakehouse pipeline on Databricks using PySpark, Delta Lake, and Unity Catalog.
- Implemented Medallion Architecture (Bronze → Silver → Gold layers) with Star Schema modeling.
- Configured incremental loading and orchestrated Databricks jobs.
- Created a live Genie AI dashboard for real-time FMCG analytics.

### ***Ball Tracking Mobile Robot***

*Computer Vision / Robotics | May 2023 | <https://github.com/Yousef-koura/Ball-Tracking-Robot>*

- Built a sensor-driven robot with a real-time vision inference pipeline.
- Achieved 95% object detection accuracy and sub-5cm tracking error.
- Implemented autonomous navigation and control algorithms.

### ***Steganography Image Detection***

*Deep Learning / Image Forensics | Feb 2023 | <https://github.com/Yousef-koura/Steganography-Detector>*

- Trained a CNN-based classifier to detect hidden data embedded in digital images.
- Achieved 90% classification accuracy.

### ***Breast Cancer AI Classifier***

*Machine Learning / Classification* | Aug 2023 | <https://github.com/Yousef-koura/Breast-Cancer-Ai>

- Built a machine learning classifier for early breast cancer detection.
- Achieved 95% accuracy.
- Awarded 3rd place at a competitive AI challenge co-hosted with the Electronics Research Institute.

### ***Machine Learning Projects Repository***

*ML Portfolio* | 2023 – ongoing | <https://github.com/Yousef-koura/Machine-Learning-Projects>

- 8+ ML projects covering regression, classification, and clustering.
- Tools: Python, Pandas, Scikit-learn, Matplotlib, NumPy.

### ***Deep Learning Projects Repository***

*DL Portfolio* | 2023 – ongoing | [https://github.com/Yousef-koura/Deep\\_Learning\\_Projects](https://github.com/Yousef-koura/Deep_Learning_Projects)

- Advanced neural network projects including CNNs, RNNs, and Transformers.
- Tools: TensorFlow, Keras, PyTorch.

### ***SQL Projects Repository***

*Data / SQL* | 2024 – ongoing | [https://github.com/Yousef-koura/SQL\\_Projects](https://github.com/Yousef-koura/SQL_Projects)

- Data analysis and management using complex SQL queries.
- Covers database design and optimization in PostgreSQL and MySQL.

### ***Chest X-Ray Pneumonia Classification (PyTorch)***

*Deep Learning / Medical Imaging* | 2025

- Comparing custom CNN versus ResNet18 transfer learning on the Kaggle COVID-19 Radiography Dataset.
- Handles class imbalance using pos\_weight in BCEWithLogitsLoss.
- Trained on GTX 1050 GPU with PyTorch 2.0.1 (CUDA 11.8).

### ***Vertical Axis Wind Turbine***

*Hardware / Renewable Energy* | April 2020

- Designed and built a vertical axis wind turbine to improve efficiency and scalability.
- Conducted research, modeled using CAD software, tested prototype, and worked on control and electrical systems.

### ***Human Following Robot***

*Robotics / Embedded Systems* | April 2021

- Designed a robot capable of following a human using sensors, microcontroller programming, and actuator integration.
- Gained hands-on experience in robotics, control systems, and embedded systems.

### ***Automatic Color Sorting Machine***

*Automation / Embedded Systems | June 2022*

- Built a machine with color sensors, servo motors, and sorting algorithms.
- Responsible for mechanical design, microcontroller programming, and sensor integration.

### ***Hot Wire Cutting CNC Machine***

*CNC / Manufacturing | Dec 2022*

- Built a CNC machine using a heated wire to cut foam, Styrofoam, and plastic.
- Covered mechanical design, CNC programming, and component integration.

## **5. Skills**

### ***Programming Languages***

Python, C++, C, MATLAB, SQL

### ***ML / Deep Learning***

Supervised Learning, Unsupervised Learning, Neural Networks (CNNs, RNNs), Transfer Learning, Object Detection (YOLOv8), LLMs, RAG, Classification, Regression, Clustering

### ***Frameworks & Libraries***

TensorFlow, PyTorch, Keras, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, FastAPI, OpenCV, LangChain

### ***Data Engineering***

ETL/ELT, Medallion Architecture, Delta Lake, PySpark, Databricks, Star Schema, Unity Catalog

### ***MLOps & DevOps***

Git, GitHub, Docker, Kubernetes, REST APIs, ONNX Runtime, AWS (fundamentals), CI/CD

### ***Hardware & Robotics***

NVIDIA Jetson Nano, Microcontrollers, ROS (Robot Operating System), PID/Fuzzy Control, Sensors & Actuators, Embedded Systems

### ***Design & CAD***

SolidWorks, AutoCAD

### ***Soft Skills***

Problem-solving, Time Management, Adaptability, Research, Leadership, Project Management, Technical Writing, Presentation

### ***Languages***

Arabic (Native), English (Professional), French (Limited working proficiency)

## 6. Publications

### ***Agri-Bot: AI-Powered Crop Disease Detection and Medication Recommendation Robot***

Published at the **8th International Undergraduate Research Conference (IUGRC 8)** — 2024

This paper presents an autonomous agricultural robot that integrates YOLOv8 for real-time crop disease detection and MobileNetV2 for classification, deployed on NVIDIA Jetson Nano. The system achieves 96% accuracy at 20 FPS, uses fuzzy PID motor control for navigation, and includes a medication recommendation module and web interface.

PDF available at: [https://github.com/Yousef-koura/yousefkoura.github.io/raw/main/AgRobot\\_paper.pdf](https://github.com/Yousef-koura/yousefkoura.github.io/raw/main/AgRobot_paper.pdf)

## 7. Certifications

- Machine Learning Specialization — Coursera (2023)
- Supervised Machine Learning: Regression and Classification — Coursera (2023)
- Python 101 for Data Science — IBM Cognitive Class (2022)
- Certified in Learning Excel: Data Analysis — LinkedIn (2022)
- Arduino Platform and C Programming — Coursera (Apr 2022)

## 8. Portfolio Website — [yousefkoura.github.io](https://yousefkoura.github.io)

Yousef maintains a personal portfolio website at [yousefkoura.github.io](https://yousefkoura.github.io), built with a black-themed, particle-animated design using Three.js and Canvas. The site includes:

- Hero section with animated role carousel: Machine Learning, Deep Learning, Researcher, Mechatronics Engineer, Data Engineer.
- Neural stats: 26 AI systems deployed, 8 certifications, 1 publication, military service completed.
- All 8 project cards with GitHub links and real project images.
- Skills grid: Programming Languages, AI & Deep Learning, MLOps, Data Engineer.
- Timeline-based experience section.
- Publication section with PDF download link.
- Contact section: email ([yousefahmed.ae20@gmail.com](mailto:yousefahmed.ae20@gmail.com)), GitHub ([github.com/Yousef-koura](https://github.com/Yousef-koura)), LinkedIn ([linkedin.com/in/yousefkoura](https://linkedin.com/in/yousefkoura)).
- AI Chatbot (YK Assistant) powered by rule-based Q&A; about Yousef's profile.
- Downloadable CV PDF hosted at GitHub.

## 9. Personal Information & Contact

**Full Name:** Yousef Ahmed Koura

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**GitHub:** [github.com/Yousef-koura](https://github.com/Yousef-koura)

**LinkedIn:** [linkedin.com/in/yousefkoura](https://linkedin.com/in/yousefkoura) / [linkedin.com/in/yousefahmedkoura](https://linkedin.com/in/yousefahmedkoura)

**Portfolio:** [yousefkoura.github.io](https://yousefkoura.github.io)

**Military Service:** Completed

**Availability:** Available for work

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End of Yousef Koura RAG Knowledge Base — Compiled from CV v1 (Yousef\_Ahmed\_CV\_final.docx), CV v2 (Yousef\_Koura\_cv.docx), and Portfolio ([yousefkoura.github.io](https://yousefkoura.github.io))